PHYS 121 — HW5 (Redesigned)

Student Name: ___________________________.   NetID: ________________.

Q1. Spider silk realism + Spiderman (16 pts)

he elastic modulus of silk is about $Y = 25$ GPa (2.5×10^{10}N/m^2). This means that a silk thread of length L and cross-sectional area A, then the tension in a silk thread stretched by an amount x would be $F = Y A\, x/L$. Consider a 50 mg spider that has a silk line with a diameter of $d = 0.14$ mm. Assume the spider is hanging from a line that is $L = 50$ cm long.

(a) By how much does the line stretch because of the weight of the spider?

(b) How much potential energy is stored in the stretched silk line?

(c) If all of this energy is released to the spider in the form of kinetic energy, what would be the speed of the spider?

In one of the Spiderman movies, Spiderman stops a falling car with a single silk line, although it is much thicker than the spider line above. Assume that the car falls out of a ten-story tall parking garage, the line is attached to the top of the parking garage, the silk line manages to connect to the car when it is only two stories above the ground, and the car comes to a rest right when it reaches the ground (similar to the vine jumpers in Vanuatu). I'm getting this info by watching a movie; the numbers are all approximate. (by Prof. Paul Stanley)

(d) Draw a picture!

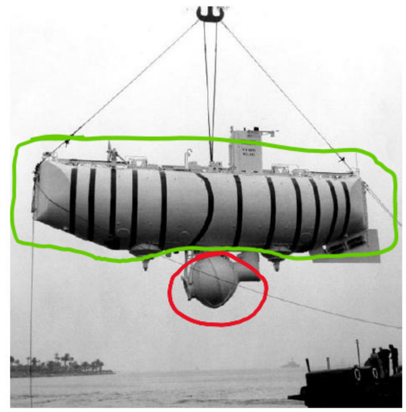
(e) Estimate the minimum necessary diameter of the silk line that Spiderman must be using for the physics to work. At this point, assume the silk is infinitely elastic.

(f) Would the line break? How do you know? At this point, don't assume the silk is infinitely elastic.   (3+3+3+1+3+3 points)

Q2. Floating ball oscillation (2+6+2 pts)

A solid sphere (radius r) floats in water (in a sink) with *half* its volume submerged. A small child pushes it down a small distance and releases. (a) What is the mass density of the ball? (b) Show the vertical motion is a simple harmonic motion (SHM) for small displacements and find the frequency f (not just ω). (c) Suppose you scale the ball down (same material, smaller r). If viscous drag matters, will the oscillation frequency increase, decrease, or stay the same as r decreases? Defend with scaling. *Required drawing: sphere at half-submergence; show cross-sectional area at the waterline and a small downward displacement y*.

Q3. Challenger Deep (3+3+3+2+2 points)



The bathyscaphe Trieste was the first to visit the deepest point of the ocean at a depth of 10,900 meters. The crew occupied a steel sphere that was 2.16 meters outside diameter.

(a) What is the difference in pressure between the inside and outside of the sphere at that depth? You can assume salt water has a density of 1,030 kg/m^3 and is incompressible, but keep your answer at three significant figures.

(b) Assuming the the sphere is divided in half, what is the magnitude of the force (because of the pressure) pushing together the two hemispheres of this sphere?

(c) The compressive yield strength of the steel is 520 MPa. Assuming that the force from the previous answer is distributed over the area of contact of the two hemispherical shells, calculate the minimum necessary thickness of the shells.

(d) The actual sphere had a thickness of 13 cm. Hopefully, your answer is somewhere close to that. Assuming a density for steel of 7,800 kg/m^3, find the mass of the spherical shell.

(e) The bathyscaphe was given neutral buoyancy by a large bag of gasoline with a density of 750 kg/m^3. Assuming all of the mass of the bathyscaphe was either gasoline or the metal spherical shell for the people, what volume of gasoline was required?

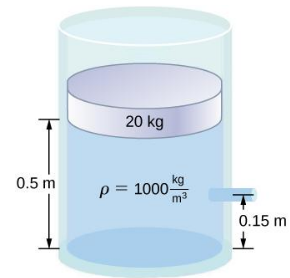
Q4. More on air pressure (2+2+4+2+2+3+0 points pts)

Assume $P(0)=P_0=100$ kPa at sea level, and $\rho_0=1.2$ kg/m³ (near sea level). (a) With constant Consider the earth's atmosphere. The air pressure at sea level is approximately P_0 = 100 kPa, while the density of air is approximately 1.2 kg/m^3. (a) Assuming that the density of the air is constant, how deep is the atmosphere? (b) In Sichuan, about 100 km south of Kangding, there is a mountain that is about 7,700 m tall. What is the air pressure at the top of this mountain, according to the previous assumption? (c) Assuming that the density of the air is proportional to the air pressure, show that the air pressure is given by $P = P_0 e^{-z/\alpha}$ by direct substitution into the differential form of Pascal's law. (d) What are the dimensions (unit) of α in the previous equation? (e) What is the numerical value of α in the previous equation? (f) What is the air pressure at the top of that Sichuan mountain, according to the variable density model? (g) What is the name of this mountain?

Q5. Pump out the water (10 pts)

Boats leak. Fortunately, there are pumps to get the water out. A certain boat is filling with water that accumulates inside at a distance of 3 meters below the outside waterline. The boat has a pump that is rated to deliver power at 2,500 Watts. (a) Assuming a pipe with a diameter of 2 cm is used to pump the water, at what rate can water be removed from the boat? (b) Assuming a pipe with a diameter of 4 cm is used to pump the water, at what rate can water be removed from the boat? (c) Which pipe size should be used, and why? (4+4+2 points)

Q6. A leaking tank with a piston (4+4+2 pts)



A tank of cross-sectional area A = 0.12 m² has a frictionless piston on top. A side spout, 0.15 m above the bottom, vents to atmosphere and ejects a horizontal jet. The spout is at height 1.50 m above the ground. The cross sectional area of the spout is $A_s = 7.0 \times 10^{-4} m^2$. (a) Using Bernoulli (surface → spout), derive the exit speed v. (b) With the jet leaving at height 1.50 m, how far horizontally does it travel before hitting the floor? (Treat the jet as a projectile. Ignore all friction and dissipative forces) (c) If the decline of the water level inside the tank is considered, how will your answers change qualitatively?

Q7. Household water supply (12 pts)

Water supplied to a house by a water main has a pressure of $3.00 \times 10^5 \mathrm{N/m^2}$ *early on a summer day* when neighborhood use is low. This pressure produces a flow of 21.0 L/min through a garden hose. *Later in the day*, pressure at the exit of the water main and entrance to the house drops, and a flow of only 8.50 L/min is obtained through the same hose. (a) What pressure is now being supplied to the house, assuming resistance is constant? (b) By what factor did the flow rate in the water main increase in order to cause this decrease in delivered pressure? The pressure at the entrance of the water main is $5.00 \times 10^5 \mathrm{N/m^2}$ which is maintained during the process, and the original flow rate was 200 L/min. (c) How many more users are there (*later in the day*), assuming each would consume 21.0 L/min *in the morning*? (4+4+4 points)

Q8. Air-speed indicator (7+3 pts)

An airplane can determine the wind speed with a device called a Venturi meter, which is a pipe with a constriction connected to a U-tube manometer. Assume the air enters the pipe with a speed V through a pipe with diameter D. The constriction has a diameter d. The fluid in the U-tube has a density ρ_w.

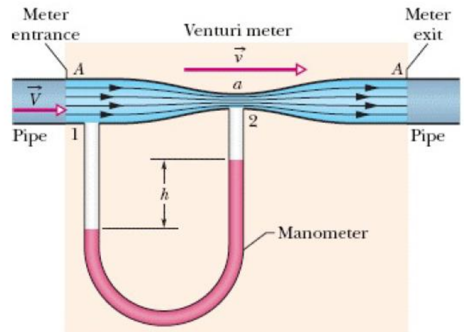


Image Credit: Stephen Zheng

(a) Assuming that air is incompressible with a constant density ρ_a, derive an expression for the velocity V in terms of the listed variables. Expect to use both the Bernoulli equation and the equation of continuity.

(b) The density of air is actually proportional to the air pressure. Explain if the true velocity V is larger, the same as, or smaller than the prediction of your formula for part (a). Defend your answer with a little physics.

Q9. Gasoline pipe (10 pts)

Gasoline is piped underground from refineries to major users. The flow rate is 3.00×10^{-2} m^3/s, the viscosity of gasoline is $1.00 \times 10^{-3}\ (\mathrm{N/m^2}) \cdot \mathrm{s}$, and its density is 680 kg/m^3. (a) What minimum diameter must the pipe have if the Reynolds number is to be less than 2000? (b) What pressure difference must be maintained along each kilometer of the pipe to maintain this flow rate? Comment on the results. (4+6 points)

Bonus (0–2 pts): Microfluidic chip

Search and briefly summarize what a typical microfluidic chip is and comment on the impact of viscosity. [less than 200 words]

Assistance notes: